Template Week 6 – Networking

Student number: 585361

Assignment 6.1: Working from home

Screenshot installation openssh-server:

```
See "man sudo_root" for details.

thije@thije-VMware-Virtual-Platform:~$ sudo apt install openssh-server
[sudo] password for thije:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  ncurses-term openssh-sftp-server ssh-import-id
Suggested packages:
  molly-guard monkeysphere ssh-askpass
The following NEW packages will be installed:
  ncurses-term openssh-server openssh-sftp-server ssh-import-id
0 upgraded, 4 newly installed, 0 to remove and 171 not upgraded.
Need to get 832 kB of archives.
After this operation, 6,743 kB of additional disk space will be used.
Do you want to continue? [Y/n] Y
Get:1 http://nl.archive.ubuntu.com/ubuntu noble-updates/main amd64 openssh-sftp-
server amd64 1:9.6p1-3ubuntu13.15 [37.3 kB]
Get:2 http://nl.archive.ubuntu.com/ubuntu noble-updates/main amd64 openssh-serve
r amd64 1:9.6p1-3ubuntu13.15 [510 kB]
Get:3 http://nl.archive.ubuntu.com/ubuntu noble/main amd64 ncurses-term all 6.4+
20240113-1ubuntu2 [275 kB]
```

```
Mar 20 11:28
thije@thije-VMware-Virtual-Platform: ~
Setting up ncurses-term (6.4+20240113-1ubuntu2) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for ufw (0.36.2-6) ...
thije@thije-VMware-Virtual-Platform:~$ sudo systemctl enable --now ssh
Synchronizing state of ssh.service with SysV service script with /usr/lib/system
d/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh
Created symlink /etc/systemd/system/sshd.service → /usr/lib/systemd/system/ssh.s
ervice.
Created symlink /etc/systemd/system/multi-user.target.wants/ssh.service → /usr/l
ib/systemd/system/ssh.service.
thije@thije-VMware-Virtual-Platform:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group defaul
t qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP gro
up default qlen 1000
    link/ether 00:0c:29:46:69:97 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
```

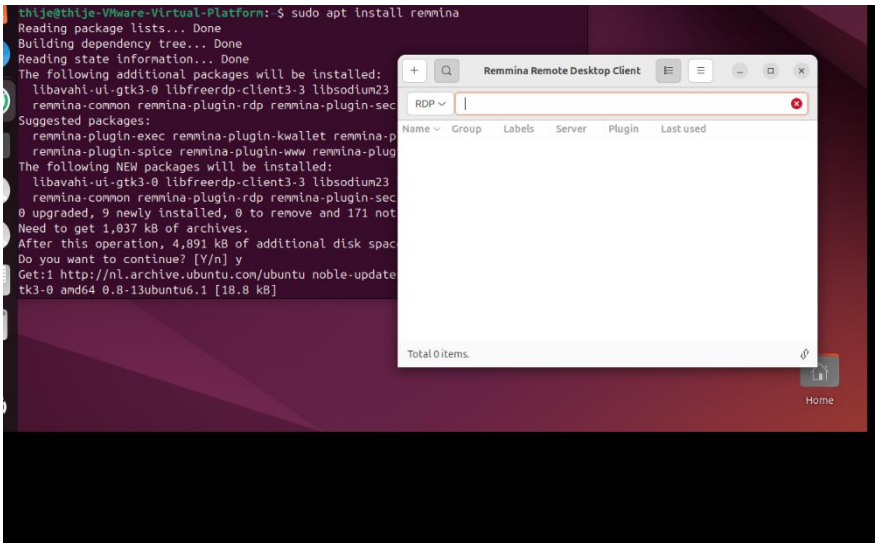
Screenshot successful SSH command execution:

```
Are you sure you want to continue connecting (yes/no/[fingerprint])
Warning: Permanently added '192.168.139.133' (ED25519) to the list o
own hosts.
thije@192.168.139.133's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.17.0-19-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

166 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

thije@thije-VMware-Virtual-Platform:~$
```

Screenshot successful execution SCP command:

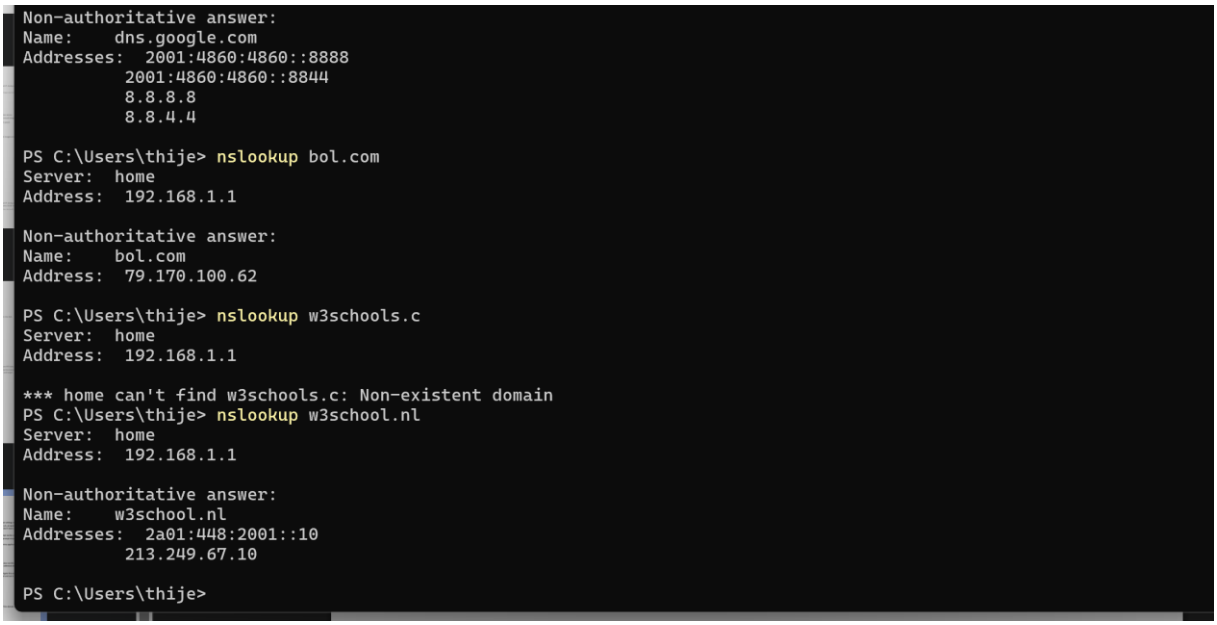
```
thije@thije-VMware-Virtual-Platform:~$ scp tst.txt thije@192.168.139.133
:/home/thije/
thije@192.168.139.133's password:
tst.txt                                   100%   19     4.5KB/s   00:00
```

Screenshot remmina:



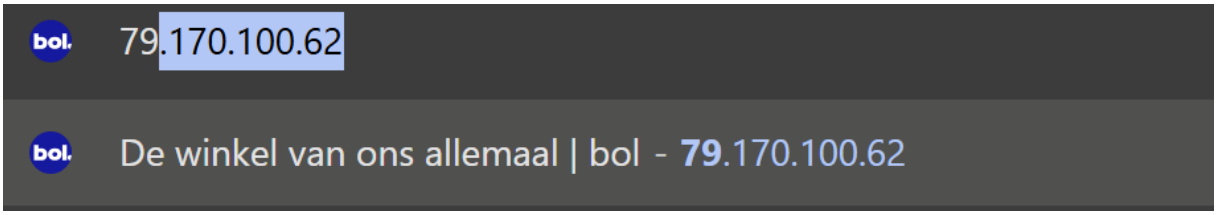
Assignment 6.2: IP addresses websites

Relevant screenshots nslookup command:



```
Non-authoritative answer:
Name:    dns.google.com
Addresses:  2001:4860:4860::8888
          2001:4860:4860::8844
          8.8.8.8
          8.8.4.4

PS C:\Users\thije> nslookup bol.com
Server:  home
Address:  192.168.1.1

Non-authoritative answer:
Name:    bol.com
Address:  79.170.100.62

PS C:\Users\thije> nslookup w3schools.c
Server:  home
Address:  192.168.1.1

*** home can't find w3schools.c: Non-existent domain
PS C:\Users\thije> nslookup w3school.nl
Server:  home
Address:  192.168.1.1

Non-authoritative answer:
Name:    w3school.nl
Addresses:  2a01:448:2001::10
          213.249.67.10

PS C:\Users\thije>
```

Screenshot website visit via IP address:



Assignment 6.3: subnetting

How many IP addresses are in this network configuration 192.168.110.128/25?

32 – 25 = 7 dus 7 bits over

2^7 = 128 – 2 = 126 over netwerknaam en broadcast

What is the usable IP range to hand out to the connected computers?

192.168.110.129 - 254

Check your two previous answers with this Linux command: `ipcalc 192.168.110.128/25`

Explain the above calculation in your own words.

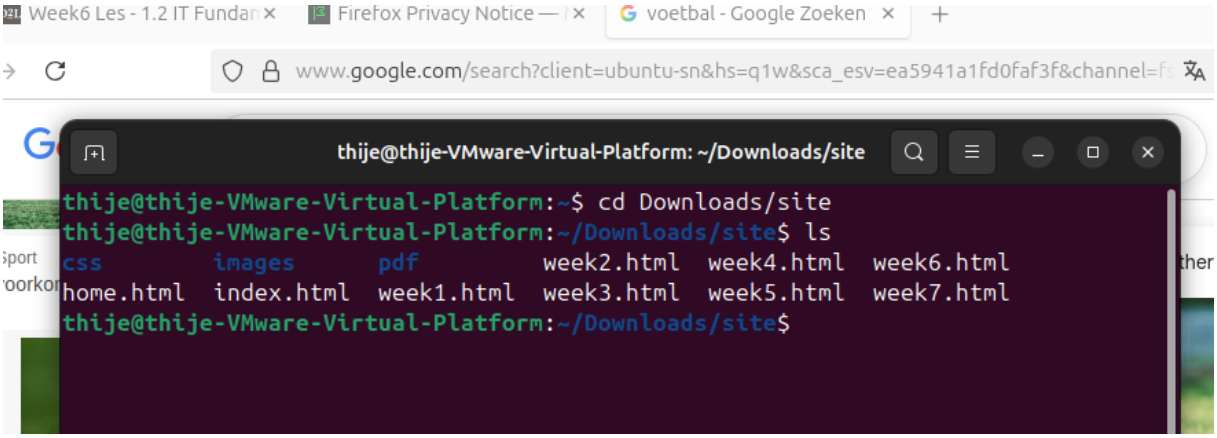
Assignment 6.4: HTML

Screenshot IP address Ubuntu VM:

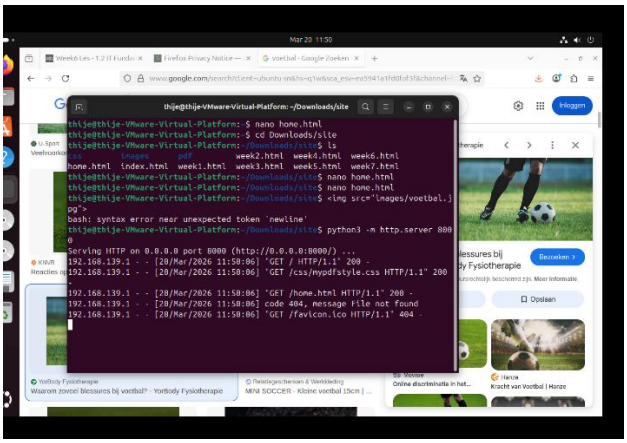
```
  link/ether 00:0c:29:46:69:97 brd ff:ff:ff:ff:ff:ff
  altname enp2s1
  inet 192.168.139.133/24 brd 192.168.139.255 scope global dynamic noprefixrou
: ens33
     valid_lft 930sec preferred_lft 930sec
```

Screenshot of Site directory contents:

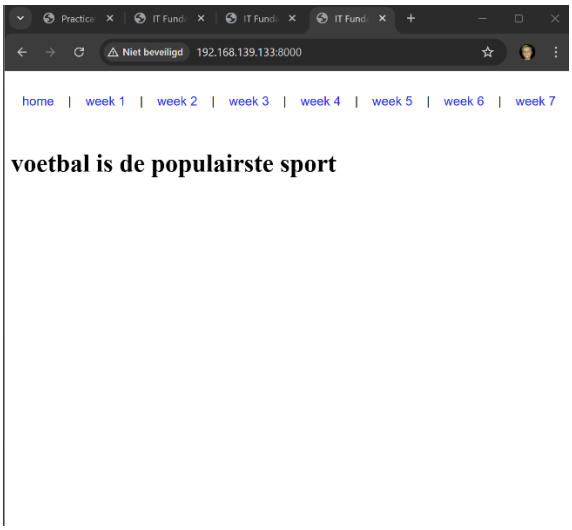
```
tml>
<head>
  <title>IT Fundamentals</title>
</head>
<body>
  <h1>voetbal is de populairste sport</h1>
</body>
html>
```



Screenshot python3 webserver command:



Screenshot web browser visits your site



Assignment 6.5: Network segment

Remember that bitwise java application you've made in week 2? Expand that application so that you can also calculate a network segment as explained in the PowerPoint slides of week 6. Use the bitwise & AND operator. You need to be able to input two Strings. An IP address and a subnet.

IP: 192.168.1.100 and subnet: 255.255.255.224 for /27

```
Example: 192.168.1.100/27
Calculate the network segment
IP Address:   11000000.10101000.00000001.01100100
Subnet Mask:  11111111.11111111.11111111.11100000
-----------------------------------------------
Network Addr: 11000000.10101000.00000001.01100000

This gives 192.168.1.96 in decimal as the network address.
For a /27 subnet, each segment (or subnet) has 32 IP addresses (2⁵).
The range of this network segment is from 192.168.1.96 to 192.168.1.127.
```

Paste source code here, with a screenshot of a working application.

Ready? Save this file and export it as a pdf file with the name: **week6.pdf**